

Table S.5

| Method                | $\alpha$ | City 1             |                    |                    |                    | City 2             |                    |                    |                    | City 3             |                    |                    |                    |
|-----------------------|----------|--------------------|--------------------|--------------------|--------------------|--------------------|--------------------|--------------------|--------------------|--------------------|--------------------|--------------------|--------------------|
|                       |          | Accuracy           | Precision          | Recall             | F <sub>0.5</sub>   | Accuracy           | Precision          | Recall             | F <sub>0.5</sub>   | Accuracy           | Precision          | Recall             | F <sub>0.5</sub>   |
| TSLE <sub>L0</sub>    | 0.15     | 93.16%<br>(0.0165) | 97.01%<br>(0.0228) | 87.96%<br>(0.0214) | 95.06%<br>(0.0225) | 89.29%<br>(0.0162) | 93.63%<br>(0.0192) | 72.66%<br>(0.0149) | 88.52%<br>(0.0181) | 93.23%<br>(0.0158) | 96.55%<br>(0.0227) | 89.77%<br>(0.0216) | 95.11%<br>(0.0224) |
|                       | 0.20     | 89.59%<br>(0.0197) | 96.87%<br>(0.0198) | 87.87%<br>(0.0184) | 94.93%<br>(0.0195) | 84.27%<br>(0.0181) | 92.86%<br>(0.0152) | 72.10%<br>(0.0117) | 87.80%<br>(0.0143) | 89.00%<br>(0.0214) | 96.80%<br>(0.0151) | 90.04%<br>(0.0142) | 95.37%<br>(0.0149) |
|                       | 0.30     | 81.09%<br>(0.0285) | 96.86%<br>(0.0119) | 87.97%<br>(0.0112) | 94.94%<br>(0.0118) | 72.50%<br>(0.0185) | 90.58%<br>(0.0148) | 70.36%<br>(0.0116) | 85.66%<br>(0.0140) | 80.57%<br>(0.0299) | 96.74%<br>(0.0096) | 89.98%<br>(0.0090) | 95.30%<br>(0.0095) |
| TSLE <sub>Lasso</sub> | 0.15     | 99.20%<br>(0.0041) | 97.36%<br>(0.0149) | 99.60%<br>(0.0028) | 97.79%<br>(0.0123) | 96.79%<br>(0.0058) | 90.94%<br>(0.0174) | 95.99%<br>(0.0060) | 91.90%<br>(0.0146) | 98.16%<br>(0.0318) | 98.17%<br>(0.0060) | 99.02%<br>(0.0020) | 98.34%<br>(0.0050) |
|                       | 0.20     | 98.39%<br>(0.0058) | 96.12%<br>(0.0155) | 99.46%<br>(0.0026) | 96.77%<br>(0.0127) | 94.03%<br>(0.0130) | 88.41%<br>(0.0203) | 94.70%<br>(0.0055) | 89.60%<br>(0.0172) | 96.65%<br>(0.0472) | 97.65%<br>(0.0058) | 98.73%<br>(0.0017) | 97.86%<br>(0.0047) |
|                       | 0.30     | 95.25%<br>(0.0155) | 93.20%<br>(0.0201) | 99.10%<br>(0.0028) | 94.32%<br>(0.0166) | 85.79%<br>(0.0227) | 83.49%<br>(0.0188) | 92.32%<br>(0.0058) | 85.11%<br>(0.0159) | 91.63%<br>(0.0825) | 96.38%<br>(0.0076) | 98.12%<br>(0.0019) | 96.73%<br>(0.0063) |
| TSLE <sub>MCP</sub>   | 0.15     | 99.65%<br>(0.0022) | 98.83%<br>(0.0081) | 99.72%<br>(0.0022) | 99.00%<br>(0.0067) | 97.02%<br>(0.0045) | 91.65%<br>(0.0150) | 95.66%<br>(0.0073) | 92.42%<br>(0.0129) | 98.93%<br>(0.0033) | 96.56%<br>(0.0120) | 98.77%<br>(0.0034) | 96.99%<br>(0.0094) |
|                       | 0.20     | 99.29%<br>(0.0033) | 98.32%<br>(0.0086) | 99.58%<br>(0.0026) | 98.57%<br>(0.0072) | 95.11%<br>(0.0058) | 90.49%<br>(0.0143) | 94.49%<br>(0.0070) | 91.26%<br>(0.0124) | 98.19%<br>(0.0051) | 95.84%<br>(0.0139) | 98.66%<br>(0.0044) | 96.38%<br>(0.0107) |
|                       | 0.30     | 98.26%<br>(0.0048) | 97.55%<br>(0.0077) | 99.39%<br>(0.0020) | 97.91%<br>(0.0064) | 89.58%<br>(0.0211) | 88.19%<br>(0.0267) | 92.90%<br>(0.0091) | 89.07%<br>(0.0213) | 96.51%<br>(0.0072) | 95.36%<br>(0.0119) | 98.36%<br>(0.0055) | 95.94%<br>(0.0089) |